

The Perturbation Catalogue:

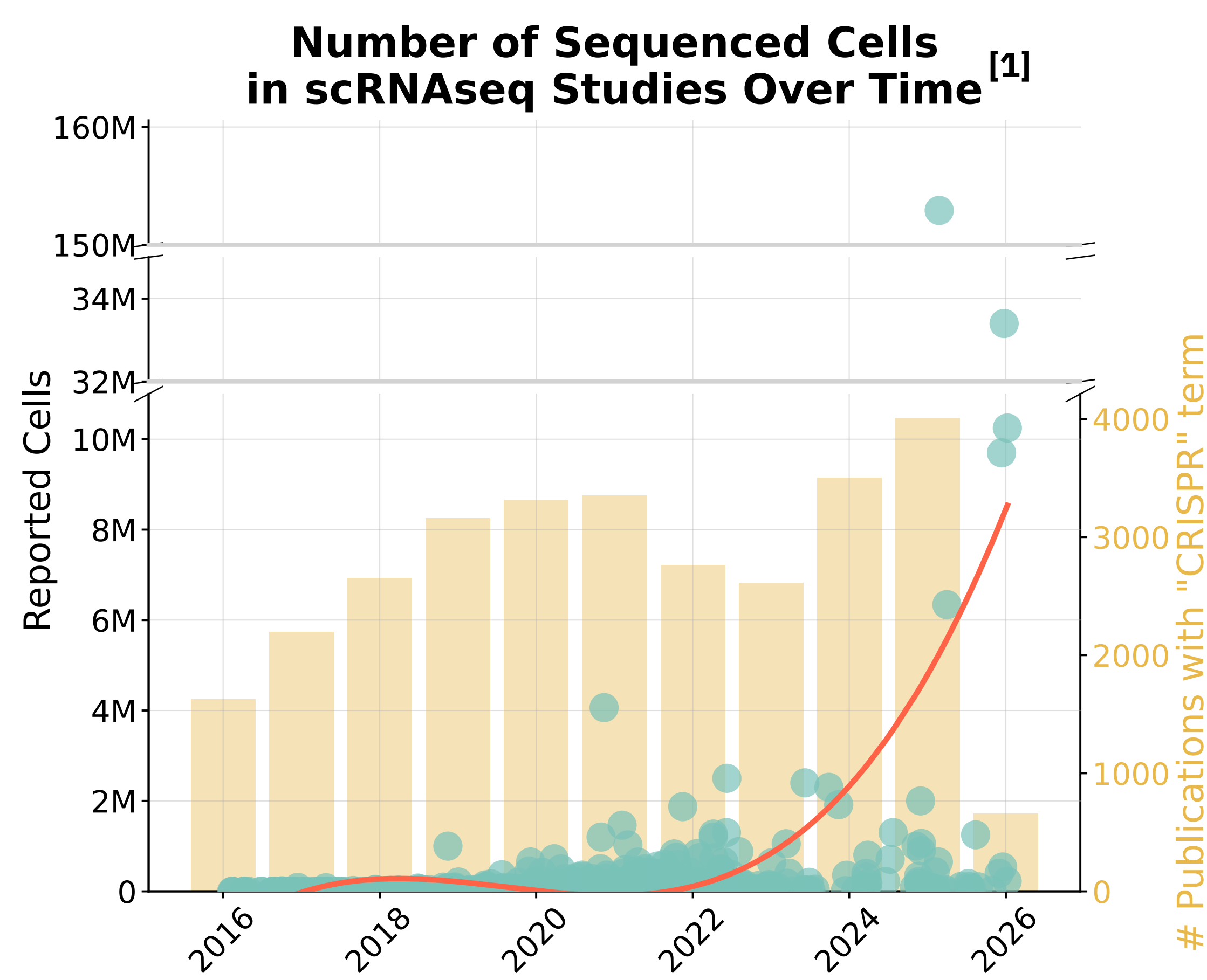
A Central Hub for Genomic Perturbation Data

EMBL-EBI

Zakirov A¹, Sokolov A¹, Tsukanov K¹, Stephenson J¹, Dunham A³, Martin MJ¹, McDonagh E¹, Uniyal P¹, Buniello A¹, Suveges D¹, Alegbe T¹, Rubin A², Fowler D⁴, Parts L³, Lotfollahi M³, Iorio F⁵, Freeberg MA¹ and industry partners

(1) European Molecular Biology Laboratory, European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, Cambridge, CB10 1SD, United Kingdom; (2) MaveDB/Walter and Eliza Hall Institute of Medical Research, Parkville VIC 3052, Australia; (3) Wellcome Sanger Institute, Wellcome Genome Campus, Hinxton, Cambridge, CB10 1SD, United Kingdom; (4) MaveDB/Department of Genome Sciences, University of Washington, Seattle, WA, 98195, USA; (5) Human Technopole, Palazzo Italia, 20157 Milan, Italy

Introduction



High-throughput genomic perturbation screening has become an essential tool in functional genomics, enabling researchers to systematically investigate gene function and interactions. Since the advent of CRISPR-Cas9 technology, there has been a steady increase in both the number of perturbation screens conducted and the scale of data generated. Meanwhile, the cost of sequencing has dramatically decreased, reaching a point where it is now economically viable to sequence tens and hundreds of millions of cells in a single screen. This has led to an explosion of data, with researchers generating vast datasets providing insights into gene function, cellular processes, and disease mechanisms.

However, the sheer volume of data generated from these screens, combined with the diversity of experimental designs and decentralised data storage, makes it difficult for the wider scientific community to access and utilise this information effectively. Moreover, the lack of standardised data formats and associated metadata standards further complicate frictionless data integration and analysis across different studies.

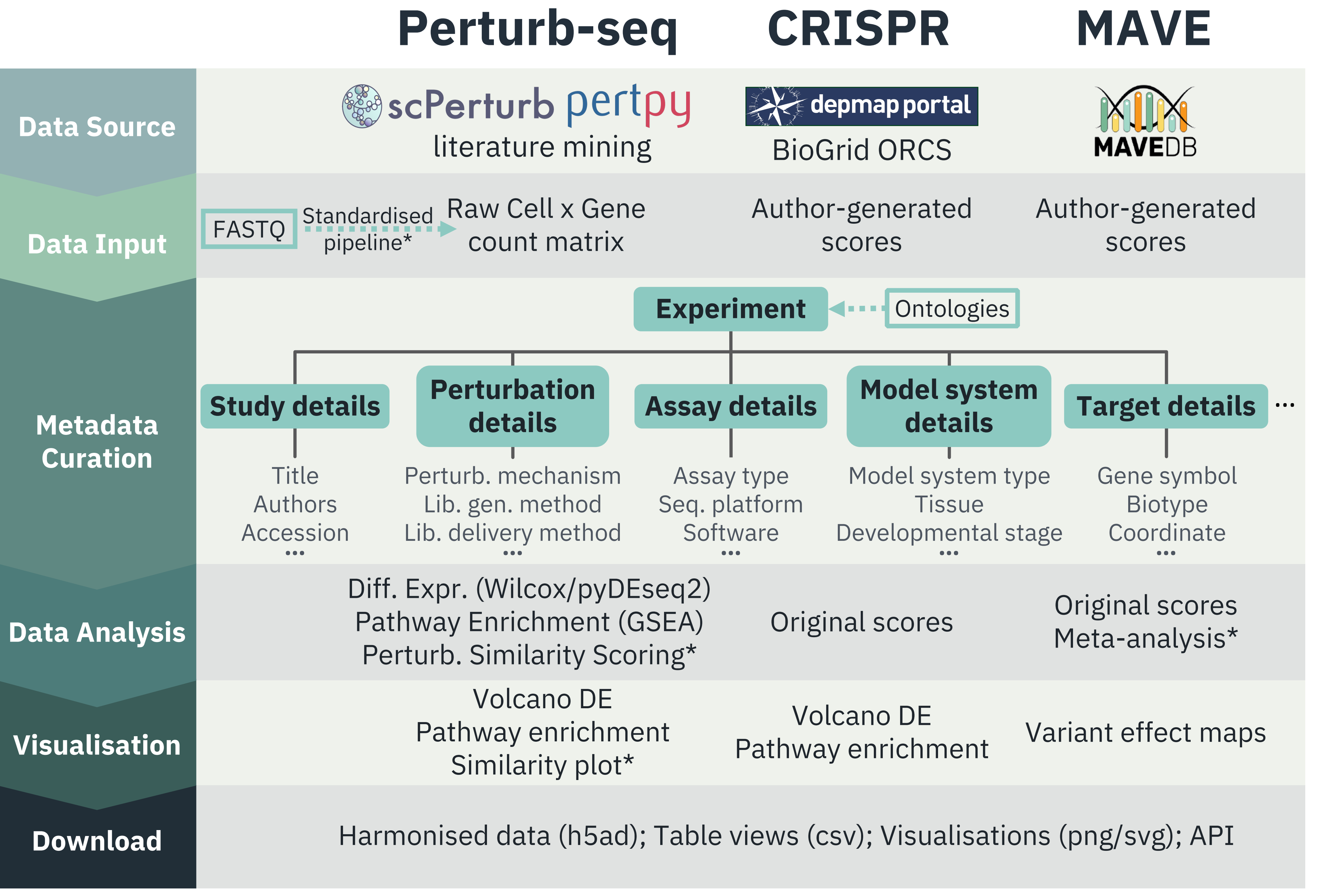
To address these issues, we have developed the Perturbation Catalogue, a centralised database that aggregates and standardises various modalities of genomic perturbation data, including Perturb-seq, CRISPR and MAVE screens. The Perturbation Catalogue provides a comprehensive resource for researchers to explore and analyse perturbation data, facilitating the discovery of novel biological insights.

Harmonisation of Diverse Genomic Perturbation Modalities

Perturbation Catalogue gathers the data from various sources, including existing public repositories (DepMap, BioGrid ORCS, MaveDB, scPerturb) and published literature. We currently preserve data in its original published form to ensure fidelity. Work is underway on a standardised pre-processing pipeline for Perturb-seq data that will further improve consistency across datasets.

The harmonisation of data from various sources is a critical step in ensuring that the perturbation data can be effectively utilised by researchers. To this end, we curate the data using a unified metadata schema that captures comprehensive information about each perturbation screen and individual perturbation, including experimental design, assay type, model system, data processing methods and more. To facilitate the consistency and discoverability of the data, we extensively use the existing ontologies and actively contribute to their further development.

Perturbation Catalogue features a user-friendly interface that allows researchers to conveniently explore and visualise the analysed data, including the differential expression, pathway enrichment and perturbation similarity. The database also provides tools for data downloading and programmatic access via API, enabling researchers to easily integrate Perturbation Catalogue data into their own workflows.



* Currently in development

Current State of The Perturbation Catalogue

Tissues

36

Cell lines

1199

Cell types

30

Perturbation data points

35.5 M

Diseases

166

Targets

19,663

Datasets

18 Perturb-seq
1197 CRISPR
10 MAVE

Conclusions

Perturbation Catalogue is a comprehensive resource that integrates diverse perturbation data modalities, enabling exploration and analysis of genetic perturbations on cellular phenotypes. Uniform data processing and harmonisation, coupled with manual data curation and extensive use of ontologies, facilitates cross-study comparisons, meta-analyses, and training of robust machine learning models.

Perturbation Catalogue freely shares knowledge with the community and acts as a focal point for high quality perturbation sets and analyses that connect targets to phenotypes. This resource will accelerate the discovery of human data-driven biological insights across various domains, including functional genomics, drug discovery, and systems biology.

[1] Valentine Svensson, Eduardo da Veiga Beltrame, Lior Pachter, A curated database reveals trends in single-cell transcriptomics, Database, Volume 2020

